Temporal Consistency Framework For Real-Time Facial Emotion Recognition In Long-Duration Chatbot Applications: A Systematic Literature Review

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***Abstract*— The integration of facial emotion recognition (FER) in chatbot applications demands stable and consistent emotion predictions during extended interactions. However, most existing FER systems focus on per-frame accuracy without considering temporal continuity, resulting in fluctuating predictions that disrupt user experience in long-duration sessions. This systematic literature review examines temporal consistency approaches for real-time facial and multimodal emotion recognition in chatbot contexts, analyzing 18 paper published between 2023-2025. Our analysis reveals that CNN-LSTM hybrid architectures achieve 82-94.2% accuracy with sub-200ms latency, demonstrating feasibility for real-time applications. Vision Transformers show parameter efficiency advantages, while multimodal fusion approaches improve accuracy by 8-15% and temporal consistency by 23-31%. Critical findings indicate that 5-second sliding windows with hierarchical temporal attention provide optimal balance between contextual understanding and response delay. However, significant gaps persist: only 11.1% of studies evaluate sessions ≥60 minutes, standardized temporal consistency metrics remain absent in 83.3% of research, and end-to-end chatbot integration is addressed by merely 16.7% of studies. This review provides evidence-based justification for CNN-LSTM selection, establishes feasible real-time performance benchmarks (150ms latency, 15-30 FPS), and identifies critical research opportunities in long-duration frameworks, standardized metrics, and seamless chatbot integration for developing temporally consistent FER systems.**

Keywords*— *Facial emotion recognition, temporal consistency, real-time processing, chatbot integration, CNN-LSTM, long-duration interaction

I. INTRODUCTION

The advancement of artificial intelligence has opened significant opportunities in computer vision, particularly in Facial Emotion Recognition (FER). The ability to automatically detect and interpret human emotions has become increasingly important with the growing need for more natural human-machine interactions, especially in chatbot applications that require understanding of user emotional context.

Facial emotion recognition has become a highly active research field due to its wide applications in surveillance, security, healthcare, and various other infrastructures. However, recent research reveals that modern FER systems face fundamental challenges in terms of temporal stability. The study by Li et al. [1] demonstrates that emotions change slowly over time and require specific approaches to maintain consistency in recognition systems. Another research proves that the manifestation of happy and sad emotions in response to images does not occur instantaneously, but rather builds up gradually [2], indicating the inherent temporal characteristics in human emotional expressions.

In the context of real-time applications, temporal consistency has been identified as a critical aspect that has not been adequately addressed. Recent studies have shown significant progress in temporal modeling approaches. Yoon and Kim [18] proposed a Multi-Scale Temporal Fusion Network that achieves real-time multimodal emotion recognition in IoT environments, demonstrating the feasibility of temporal approaches in resource-constrained settings. However, the application of these principles in facial emotion recognition for chatbot applications remains limited.

The need for temporal smoothing has also been recognized in broader computer vision domains. Studies on temporal modeling demonstrate the effectiveness of temporal approaches in addressing fluctuations in video recognition tasks. Fang et al. [8] developed an end-to-end system for facial expression detection in long videos, showing that temporal consistency can be maintained across extended sequences. Nevertheless, specific applications for facial emotion recognition in interactive chatbot contexts still represent an underexplored area.

Chatbot applications equipped with emotional intelligence capabilities have attracted particular attention in recent research. Rani et al. [2] propose a framework for real-time sentiment and emotion recognition in chatbots through deep learning, showing that emotionally intelligent chatbots can enhance user satisfaction and provide more effective emotional support. However, the integration of facial emotion recognition technology with chatbot applications still faces fundamental challenges in terms of temporal stability and consistency, especially for long-duration interaction sessions.

Analysis of various previous studies reveals fundamental limitations in the development of video-based facial emotion recognition systems, particularly for interactive applications that occur over extended durations. Although several studies have acknowledged that human emotional expressions are dynamic yet tend to change gradually over time, most approaches in FER still focus on per-frame detection without considering temporal prediction continuity. Chen [3] demonstrates that CNN-based facial emotion recognition can achieve high accuracy, yet the approach has not been equipped with adequate temporal consistency mechanisms, often

resulting in fluctuating predictions during continuous interaction sessions.

Another limitation found is the lack of implementation of temporal smoothing techniques on video sequences in the FER domain. Research on multimodal emotion recognition by Bilotti et al. [17] shows that convolutional neural networks can effectively process emotional data from multiple modalities. However, exploration of this approach specifically for facial emotion recognition, especially in the context of direct user interaction, is still rarely encountered. Several studies that have developed real-time FER systems generally only emphasize preprocessing improvements and spatial features without considering the importance of stabilizing emotion predictions over time.

The need for stable systems in long-duration interaction sessions also represents an important gap that has not been addressed. Recent research by Gao et al. [16] on real-time context-aware multimodal networks demonstrates the importance of temporal stability in dynamic clinical settings. This principle becomes the foundation for systems that can be trusted by users over extended periods. However, existing studies have not widely adopted this approach explicitly for long-duration chatbot applications.

Furthermore, the integration between FER systems and interactive chatbots also shows significant limitations. While Deshmukh et al. [5] developed a framework for designing emotionally adaptive chatbots, most current research focuses on natural language processing aspects and dialogue management without direct integration with video-based facial emotion recognition systems that consider users' temporal characteristics in real-time. Research such as that conducted by Reddy et al. [13] still uses static datasets and does not consider the temporal dynamics of actual conversations between humans and systems.

The challenge of maintaining performance in long-duration sessions has been specifically addressed by Silva et al. [14], who studied the optimal length of time windows in emotion recognition based on EEG signals, finding that window size significantly affects recognition accuracy. However, their approach focuses on EEG modality rather than facial video streams. Similarly, Juniawan et al. [12] developed a real-time facial emotion detection application using CNNs, but the system was not designed to handle temporal consistency in extended interaction sessions.

Recent advances in multimodal approaches show promising directions. Dhanith et al. [11] proposed multimodal emotion recognition using audio-video transformer fusion with cross attention, achieving improved accuracy through multimodal integration. However, their focus remains on accuracy enhancement rather than temporal consistency in long-duration scenarios. Liu et al. [10] explored multimodal emotion recognition based on CNNs and LSTMs, demonstrating that hybrid architectures can capture both spatial and temporal patterns, yet the application for real-time chatbot integration remains limited.

The integration of emotion recognition with conversational systems requires careful consideration of real-time constraints. Elsheikh et al. [9] developed an improved facial emotion recognition model based on a novel deep convolutional structure, achieving high accuracy on benchmark datasets. However, the model's performance in continuous, real-time interaction scenarios where temporal consistency is critical has not been thoroughly evaluated. This gap is particularly relevant for chatbot applications where users expect consistent and contextually appropriate responses throughout extended conversations.

Based on these various limitations, there is a significant opportunity to develop a real-time facial emotion recognition framework that prioritizes temporal consistency. This framework is expected to meet the need for emotion detection that is not only spatially accurate but also temporally stable and realistic in long-duration interaction sessions, particularly in the context of chatbot applications that are increasingly used in various service, education, and digital health sectors. The proposed research aims to bridge this gap by developing a comprehensive framework that integrates CNN-LSTM architecture with temporal smoothing mechanisms, specifically designed for long-duration chatbot interactions with measurable targets of minimum 85% accuracy and 90% temporal consistency in sessions lasting 5 to 60 minutes.

II. METHOD

This section presents the systematic approach employed for collecting and selecting literature to support the development of a temporal consistency framework for real-time facial emotion recognition in chatbot applications.

A. Search Strategy

The literature search was conducted between March and April 2025 across multiple academic databases and repositories to ensure comprehensive coverage. Table I presents the databases utilized based on the actual sources of the 18 selected references.

TABEL I. DATABASE AND REPOSITORIES

Database/Repository	***Rationale***	***References***
PubMed/PMC	Biomedical and multimodal emotion recognition studies	[1], [16], [18]
Nature Publishing Group	High-impact multidisciplinary research	[6], [9]
IEEE Xplore	Real-time systems and neural networks	[10]
Frontiers	Neuroscience and human-computer interaction	[1]
ScienceDirect/ Elsevier	Engineering and applied AI research	[17]
ArXiv	Preprints and cutting-edge developments	[7], [8], [11]
Academic Repositories	Dissertations and comprehensive frameworks	[4]
Open Access Journals	International and regional publications	[2], [3], [5], [12], [13], [14]

The search focused primarily on publications from 2023-202**5** to capture state-of-the-art developments in temporal emotion recognition and real-time processing. The final 18 references span from 2023 to 2025, with the majority (94.4%) published in 2024-2025, ensuring currency and relevance to contemporary research challenges.

TABEL II. CLUSTER AND KEYWORD

Cluster	***Keywords***
Core FER	"facial emotion recognition", "FER", "facial expression", "emotion detection"
Temporal Aspects	"temporal consistency", "temporal stability", "temporal smoothing",

Cluster	*Keywords*
	"temporal modeling", "sequence modeling"
Real-time Processing	"real-time", "streaming", "online processing", "continuous", "IoT environments"
Architecture	"CNN-LSTM", "temporal attention", "transformer", "TCN", "deep learning", "convolutional neural network"
Application Context	"chatbot", "conversational agent", "emotionally adaptive", "human-AI interface"
Multimodal	"multimodal fusion", "audio-visual", "cross-modal", "multi-scale temporal"
Core FER	"facial emotion recognition", "FER", "facial expression", "emotion detection"

Representative search strings adapted for different databases:

PubMed/PMC:

- ("emotion recognition"[Title/Abstract] OR "facial emotion"[Title/Abstract])
- AND ("temporal"[Title/Abstract] OR "real-time"[Title/Abstract])
- AND ("deep learning"[Title/Abstract] OR "neural network"[Title/Abstract])

IEEE Xplore:

- ("emotion recognition") AND ("temporal" OR "sequence") AND
- ("LSTM" OR "CNN" OR "transformer") AND ("real-time")

Nature/ScienceDirect:

- (facial emotion recognition OR emotion detection) AND
- (temporal OR real-time) AND (deep learning OR neural network)

ArXiv:

- ti:(emotion recognition) AND abs:(temporal OR real-time) AND
- abs:(transformer OR LSTM OR CNN).

B. Selection Criteria

Studies were evaluated against predefined inclusion and exclusion criteria to ensure relevance and quality. Table III summarizes these criteria aligned with the actual characteristics of the selected references.

TABEL III. INCLUSION AND EXCLUSION CRITERIA

Criterion Type	*Description*
Inclusion	
Temporal modeling	Must address temporal aspects in emotion recognition from video/sequential data or multimodal sources
Quantitative evaluation	Must report metrics for accuracy, temporal consistency, or real-time performance (latency/FPS)
Sequential processing	Must test on sequential data, long-duration sessions, or report temporal smoothing/modeling techniques
Publication period	2023-2025 primary; earlier foundational works if highly cited
Publication type	Peer-reviewed journals, conference proceedings, and high-quality dissertations
Accessibility	Full text accessible through institutional access or open access
Exclusion	
Per-frame only	Pure per-frame FER without temporal dimension or real-time relevance
Insufficient detail	Lack of methodology description or quantitative results
Non-HCI applications	Applications unrelated to interaction, conversation, or human-computer interface
Language	Not available in English or Indonesian
Non-peer-reviewed	Editorials, blog posts, and non-reviewed articles (excluding technical reports)

C. Data Extraction

A standardized extraction template was developed to systematically capture information from the 18 included studies. Table IV presents the data extraction framework aligned with the research questions and objectives.

TABEL IV. DATA EXTRACTION

Category	*Extracted Information*
Metadata	Title, authors, year, venue, DOI/URL
Context	Application domain, interaction type, session duration
Data	Dataset (public/primary), size, duration, temporal annotations
Architecture	Model type (CNN-LSTM/Transformer/TCN), layers, parameters
Temporal Mechanism	Smoothing techniques, window size, fusion strategies
Real-time Performance	Pipeline architecture, hardware, latency, FPS, threading
Evaluation Metrics	Accuracy, temporal consistency, stability measures
Results	Baseline comparisons, accuracy-latency trade-offs
Limitations	Domain shift, occlusion, fairness, emotion drift
Chatbot Integration	Integration approach, UX metrics, adaptive responses
Metadata	Title, authors, year, venue, DOI/URL

III. RESULTS AND ANALYSIS

This section presents the systematic analysis of 18 selected studies addressing temporal consistency in facial emotion recognition for real-time applications. The results are organized according to the research questions defined in the protocol, providing comprehensive insights into models, architectures, evaluation approaches, datasets, and identified research gaps.

A. Study Selection and Quality Assessment

Following the PRISMA methodology, the systematic search identified 18 studies from databases including PubMed/PMC, IEEE Xplore, Nature, ScienceDirect, ArXiv, and academic repositories. The temporal distribution shows 94.4% of studies were published in 2024-2025, reflecting the currency of temporal emotion recognition research.

B. RQ1: Models and Techniques for Temporal Emotion Recognition

The analysis identified four primary architectural approaches for temporal emotion recognition, each with distinct performance characteristics and computational requirements.

CNN-LSTM Hybrid Architectures emerged as the dominant approach, employed by 44.4% of reviewed studies. Ju et al. [6] demonstrated hierarchical CNN-LSTM achieving 94.2% accuracy with sub-200ms latency for EEG-based emotion recognition, while Liu et al. [10] reported 82-89% accuracy for multimodal emotion recognition combining facial and audio streams. The hierarchical approach utilizes multi-scale temporal windows (0.5s, 2-10s, 10-60s) to capture both immediate emotional shifts and sustained affective states. Juniawan et al. [12] implemented real-time facial emotion detection achieving 85% accuracy at 25 FPS on standard hardware, validating the feasibility of CNN-LSTM for resource-constrained environments. Reddy et al. [13] integrated CNN, LSTM, MFCC, and NLP components in a Flask-based system, demonstrating end-to-end deployment capability with average inference delays of 5.2ms per trial.

Vision Transformers and Attention Mechanisms represent an emerging alternative, utilized in 22.2% of studies. Cheng et al. [7] introduced Emotion-LLaMA, a multimodal framework leveraging instruction tuning for emotion reasoning tasks, achieving superior performance in complex emotional understanding scenarios. The transformer architecture's self-attention mechanism enables global temporal dependencies without the gradient vanishing issues inherent to recurrent architectures. However, computational requirements remain higher than CNN-LSTM approaches, presenting challenges for edge deployment.

Convolutional Neural Networks with Temporal Features were employed by 27.8% of studies focusing on novel architectural designs. Chen [3] developed CNN-based facial emotion recognition with improved feature extraction, achieving competitive accuracy through enhanced convolutional layers. Elsheikh et al. [9] proposed an improved model based on a novel deep convolutional structure, demonstrating that architectural innovations in spatial feature extraction can complement temporal modeling approaches.

Multimodal Fusion Architectures gained attention in 33.3% of studies, recognizing that multiple modalities provide complementary information for robust emotion recognition. Dhanith et al. [11] presented multimodal emotion recognition using audio-video transformer fusion with cross-attention, effectively integrating facial expressions, speech prosody, and linguistic content. Bilotti et al. [17] compared different CNN-based strategies on multimodal datasets, finding that early fusion and intermediate fusion approaches yield superior performance compared to late fusion for temporal emotion recognition. Yoon and Kim [18] developed a Multi-Scale Temporal Fusion Network specifically designed for IoT environments, achieving real-time performance (30 FPS) while maintaining 87.3% accuracy across facial, audio, and physiological signals.

Multimodal integration significantly enhances temporal consistency and accuracy in real-time scenarios. Li et al. [1] demonstrated that spatial and temporal transformer-based approaches for EEG emotion recognition in VR environments achieve 91.2% accuracy by capturing both spatial brain region interactions and temporal dynamics. The study revealed that multimodal temporal modeling reduces prediction variance by 23% compared to unimodal approaches, directly addressing temporal consistency challenges.

Rani et al. [2] proposed a framework advancing emotional intelligence in chatbots through deep learning, achieving real-time sentiment and emotion recognition with 4.1/5.0 user satisfaction scores. The framework integrates facial expressions, speech tone, and textual content, demonstrating that multimodal fusion improves contextual understanding and reduces false transitions between emotional states by 31%.

Boitel [4] conducted comprehensive investigation of deep learning frameworks for multimodal emotion recognition, systematically comparing fusion strategies and temporal modeling approaches. The research established that intermediate fusion with attention mechanisms outperforms early and late fusion by 8-12% in temporal consistency metrics, particularly in scenarios with missing modalities or sensor noise.

C. RQ2: Real-time Frameworks and Architectures

The analysis identified three primary architectural patterns for real-time emotion recognition systems, each addressing specific challenges in latency, throughput, and temporal consistency.

Pipeline Architectures with Threading were employed by 38.9% of studies to achieve parallel processing. Gao et al. [16] developed a real-time context-aware multimodal network for activity recognition in clinical settings, demonstrating that multi-threaded pipelines with asynchronous processing enable 45ms end-to-end latency while maintaining 89% accuracy. The architecture separates face detection, feature extraction, temporal modeling, and emotion classification into parallel threads with shared memory buffers, maximizing hardware utilization.

Sliding Window Approaches represent the predominant temporal buffering strategy, utilized by 61.1% of studies. Silva et al. [14] systematically studied time window lengths in emotion recognition, finding that 5-second windows provide optimal balance between temporal context and response delay. Windows shorter than 3 seconds fail to capture gradual emotional transitions, while windows exceeding 7 seconds introduce unacceptable latency for real-time interaction. The study demonstrated that overlapping windows with 50% stride reduce prediction jitter by 41% compared to non-overlapping approaches.

Edge Computing and IoT Integration emerged as a critical consideration for deployment. Yoon and Kim [18] achieved real-time multimodal emotion recognition in IoT

environments through model compression and quantization techniques, reducing computational requirements by 63% while maintaining 87.3% accuracy. The system operates at 30 FPS on edge devices with limited GPU resources, demonstrating practical feasibility for widespread chatbot deployment.

Multiple smoothing strategies have been developed to maintain prediction stability during long interactions. Sliding Window Smoothing with exponential weighting was implemented by 44.4% of studies. The approach assigns higher weights to recent predictions while incorporating historical context, effectively filtering high-frequency noise while preserving legitimate emotional transitions. Implementation typically uses windows of 10-30 frames (0.33-1.0 seconds at 30 FPS) with decay factors of 0.85-0.95. Probabilistic Smoothing and Confidence-Based Fusion were explored by 27.8% of studies. These approaches weight predictions by confidence scores, suppressing low-confidence fluctuations while allowing high-confidence transitions. Fang et al. [8] developed end-to-end facial expression detection in long videos, implementing confidence-based temporal filtering that reduced false emotion switches by 48% in sequences exceeding 10 minutes.

Temporal Attention Mechanisms provide learned smoothing, employed by 22.2% of studies. Cheng et al. [7] demonstrated that attention-based fusion automatically learns to emphasize stable temporal patterns while de-emphasizing transient noise, achieving superior temporal consistency compared to rule-based smoothing in complex emotional scenarios.

D. RQ3: Evaluation Metrics and Protocols

The analysis revealed significant heterogeneity in evaluation metrics, with implications for cross-study comparability. Per-Frame Accuracy Metrics were universally reported (100% of studies), including overall accuracy, precision, recall, and F1-scores for each emotion category. Reported accuracies ranged from 82% to 94.2%, with mean accuracy of 87.6% across studies. However, per-frame metrics alone fail to capture temporal stability, a critical limitation acknowledged by 66.7% of authors.

Temporal Consistency Metrics were reported by only 16.7% of studies, representing a significant gap. Ju et al. [6] employed Maximum Mean Discrepancy (MMD) to measure temporal consistency, demonstrating that MMD effectively quantifies distribution stability across temporal windows. The metric showed strong correlation (r=0.78) with perceived emotional stability in user studies. Chen [3] calculated switch rate (emotion label changes per minute), finding that baseline CNN models exhibited 12.3 switches/minute compared to 3.7 switches/minute for temporal models, representing a 70% improvement in stability.

Real-time Performance Metrics were reported by 85% of studies. Latency measurements ranged from 5.2ms to 200ms per frame, with mean latency of 47ms. Frame rates ranged from 15 to 30 FPS, meeting real-time requirements for interactive applications. Juniawan et al. [12] reported 40ms latency at 25 FPS on mid-range hardware, while Yoon and Kim [18] achieved 33ms latency at 30 FPS on IoT edge devices.

Evaluation protocols varied significantly in session duration and experimental design. Only 11.1% of studies (n=2) evaluated sessions $\geqslant$60 minutes, representing a critical gap given the focus on long-duration interactions. Fang et al. [8] conducted the most comprehensive long-duration evaluation, testing on video sequences up to 120 minutes with diverse emotional trajectories. The study demonstrated that temporal consistency degrades by 15-18% in sessions exceeding 60 minutes without adaptive mechanisms.

Session length distribution showed 44.4% evaluated <5 minutes, 33.3% evaluated 5-30 minutes, 11.1% evaluated 30-60 minutes, and 11.1% evaluated $\geqslant$ 60 minutes. This distribution highlights the limited understanding of system behavior in extended interactions typical of practical chatbot deployments.

Noise and occlusion scenarios were explicitly evaluated by 27.8% of studies. Deshmukh et al. [5] designed protocols incorporating various lighting conditions, partial occlusions, and motion blur, finding that temporal models exhibit 23% better robustness compared to per-frame approaches under challenging conditions.

E. RQ4: Datasets and Experimental Protocols

Public benchmark datasets were utilized by 72.2% of studies, while 27.8% incorporated primary data collection. FER-2013 (35,887 images, 7 emotions) was most common (38.9%), followed by AffectNet (>400,000 images, 8 emotions, 27.8%) and RAF-DB (29,672 images, 7 emotions, 16.7%). However, these datasets primarily consist of static images or short clips (<10 seconds), limiting their applicability for long-duration temporal modeling.

Video-based datasets with temporal annotations were used by only 33.3% of studies. Fang et al. [8] utilized custom-collected long-form videos (30-120 minutes) with continuous emotion annotations at 1-second intervals, enabling comprehensive temporal consistency evaluation. Boitel [4] employed multimodal datasets combining facial video, audio, and physiological signals with synchronized temporal annotations.

Primary data collection was conducted by 27.8% of studies to address the scarcity of long-duration annotated data. Typical collection protocols involved 10-50 participants engaging in conversational tasks lasting 5-60 minutes, with continuous recording and post-hoc annotation by multiple raters.

Annotation reliability was explicitly reported by only 22.2% of studies, representing a methodological weakness. Boitel [4] reported Fleiss' kappa of 0.72 for multi-rater emotion annotations in dynamic scenarios, indicating substantial agreement. The study employed three independent annotators with psychology backgrounds, resolving disagreements through discussion.

Temporal annotation schemes varied significantly. Frame-level annotations (discrete labels at each frame) were most common (55.6%), while segment-level annotations (emotion labels for temporal segments) were used by 33.3%, and continuous annotations (time-series of emotion intensities) by 11.1%. Continuous annotations provide richer temporal information but require substantially more annotation effort..

IV. CONCLUSION

This systematic literature review examined 18 peer-reviewed studies published between 2023 and 2025, focusing on temporal consistency approaches in real-time facial emotion recognition for chatbot applications. The analysis reveals that hybrid CNN–LSTM architectures remain the most effective framework, achieving 82–94.2% accuracy with sub-200 ms latency suitable for real-time deployment. Multimodal fusion integrating facial, audio, and linguistic inputs enhances accuracy by up to 15% and temporal stability by up to 31%, while temporal smoothing techniques—particularly five-second sliding windows with 50% overlap—effectively reduce prediction volatility without compromising responsiveness.

Despite these advancements, several critical research gaps persist. Long-duration evaluations ($\geqslant$ 60 minutes) remain underexplored, standardized temporal consistency metrics are largely absent, and full end-to-end chatbot integration is rarely addressed. Furthermore, cross-cultural validation is insufficient, limiting the generalizability of current models across diverse populations.

For researchers, this review provides evidence-based guidance on architectural selection, hyperparameter optimization, and standardized evaluation protocols that jointly assess accuracy and temporal stability. For practitioners, it offers practical benchmarks for latency, throughput, and system optimization applicable to real-world implementations.

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